Universe Routing & Latency Mathematical Reference

1. Void Distance (L)

$$L = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \times S - (R_1 + h_1) - (R_2 + h_2)$$

Symbol Definitions:

- x_1, y_1 / x_2, y_2: Coordinates of the origin / destination planet centers, as specified in the configuration.
- S: `coordinate_scale_unit_km` (from `universe_metadata`) — converts grid units to km.
- R_1 / R_2: `radius_km` of the origin / destination planet.
- h_1 / h_2: `atmosphere_thickness_km` of the origin / destination planet.

2. Void Travel Time (T_v)

$$T_v = \frac{(h_1 \times n_1) + (h_2 \times n_2) + L}{C}$$

Symbol Definitions:

- h_1 / h_2: `atmosphere_thickness_km` of the origin / destination planet.
- n_1 / n_2: `refraction_index` of the origin / destination planet.
- L: Void distance from Formula 1.
- C: `speed_of_light_kms` (from `universe_metadata`; defaults to `300,000` km/s).

3. Internal Crust Transit Time (T_p)

$$T_p = \frac{2\pi r \times s}{N \times f \times C} + m \times \Delta t$$

Symbol Definitions:

- r: `radius_km` of the current planet.
- N: `active_towers` of the current planet (total towers on the ring).
- s: Number of segments traveled along the ring between entry and exit tower (angular distance $\div\ 360^\circ / N$). Note that $s = 0$ if entry tower = exit tower.

- m: Number of distinct towers hit (for the processing-delay charge). $m = s + 1$ in general; $m = 1$ when entry tower = exit tower (the dedup case).
- f: `fiber_speed_fraction` (from `universe_metadata`; defaults to `0.67`).
- C: `speed_of_light_kms` (same constant as in T_v).
- Δt: `tower_processing_delay_ms` (from `universe_metadata`; defaults to `7` ms).

End-to-End Route Composition

$$\text{Total Latency} = \sum_{i=1}^{k} T_p(P_i) + \sum_{i=1}^{k-1} T_v(P_i, P_{i+1})$$

Core Rules:

- **One** T_p per planet visited (handling internal routing and tower delay).
- **One** T_v per void hop between consecutive planets.
- **No double-counting**, and $t/\Delta t$ only ever enters through the $m \times \Delta t$ term inside T_p.